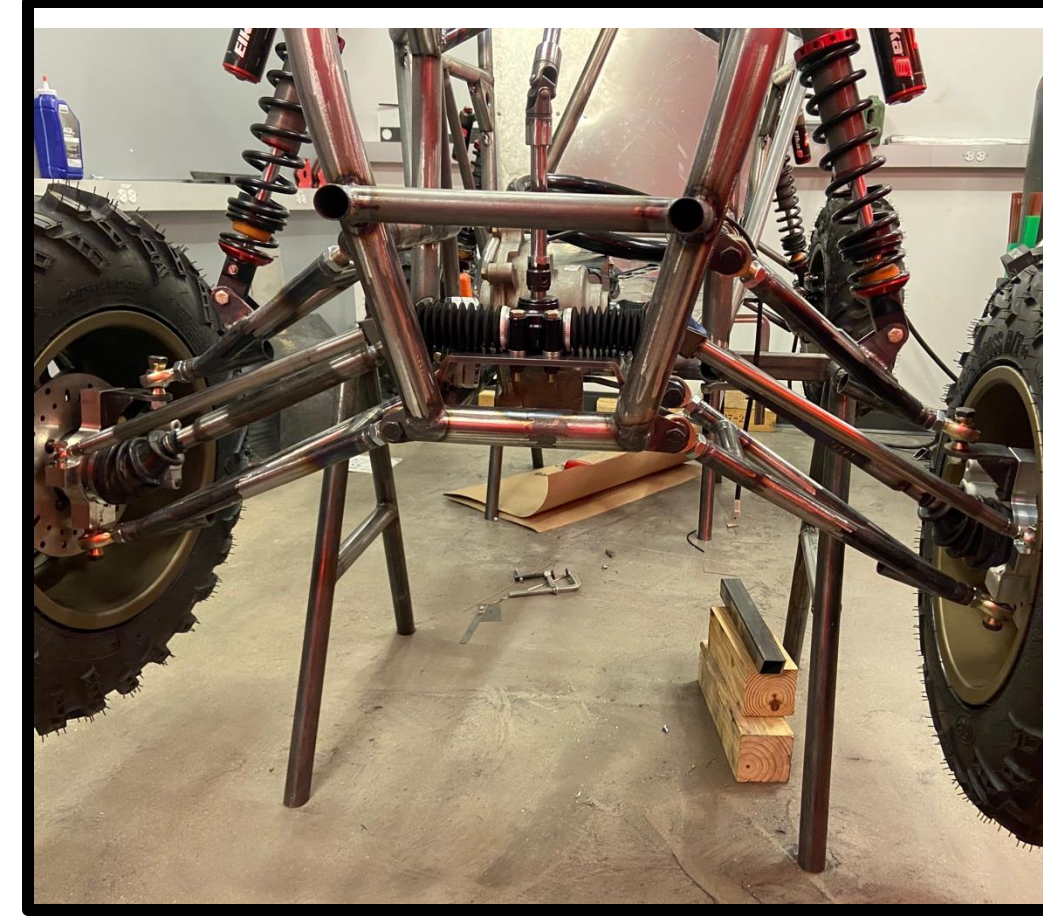


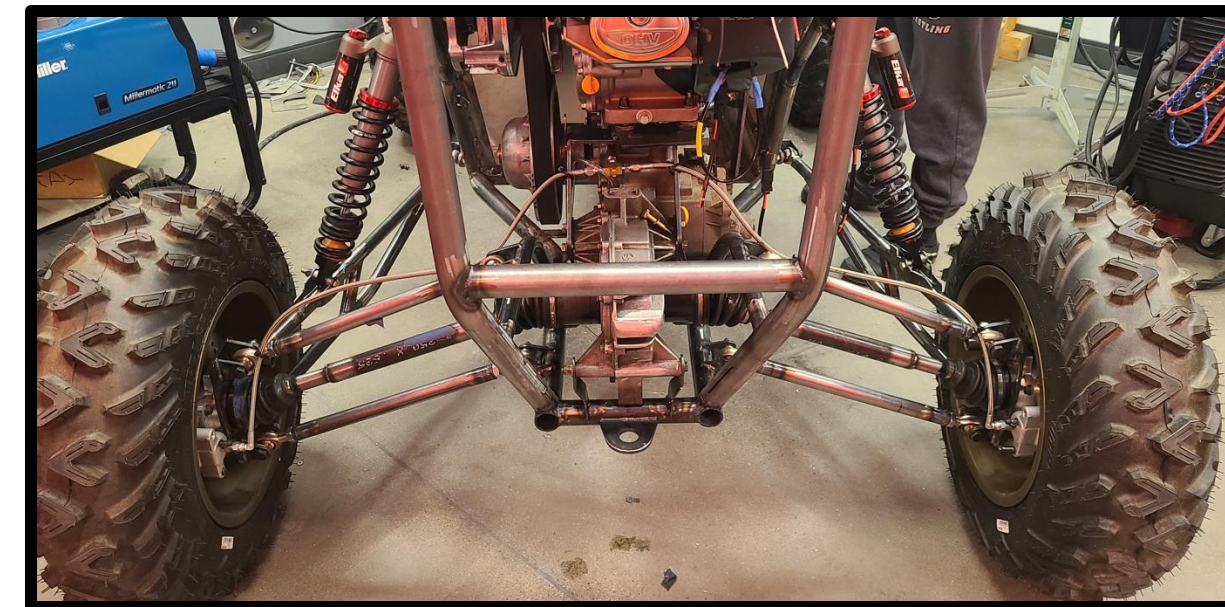
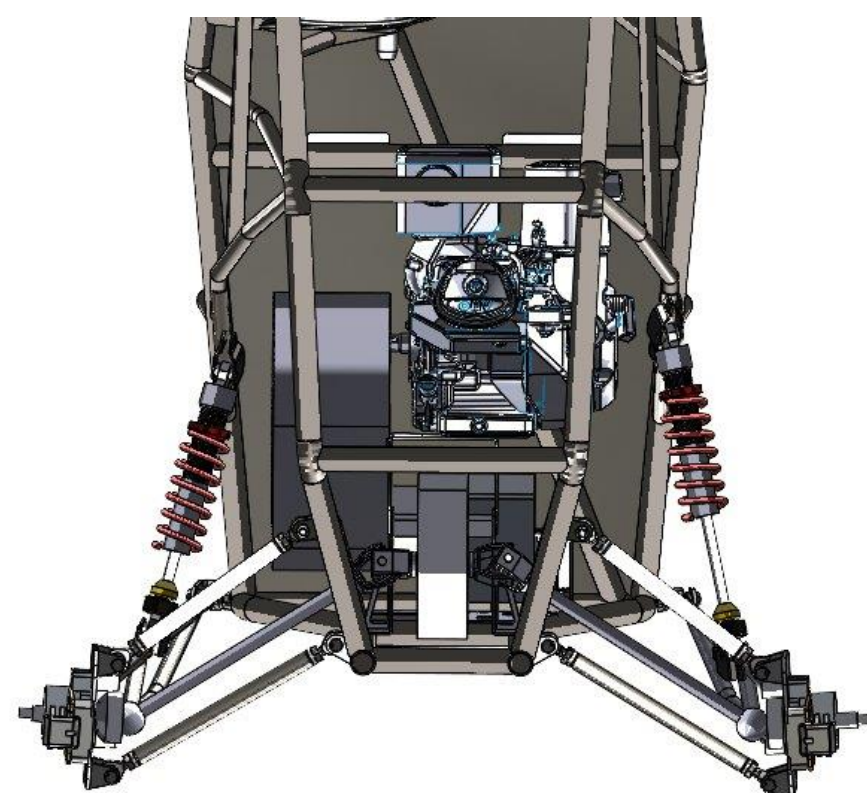
Design

Suspension:

- Elka Stage 4 Suspension on all 4 corners of the vehicle
- Low-speed compression, rebound settings, and preload adjust



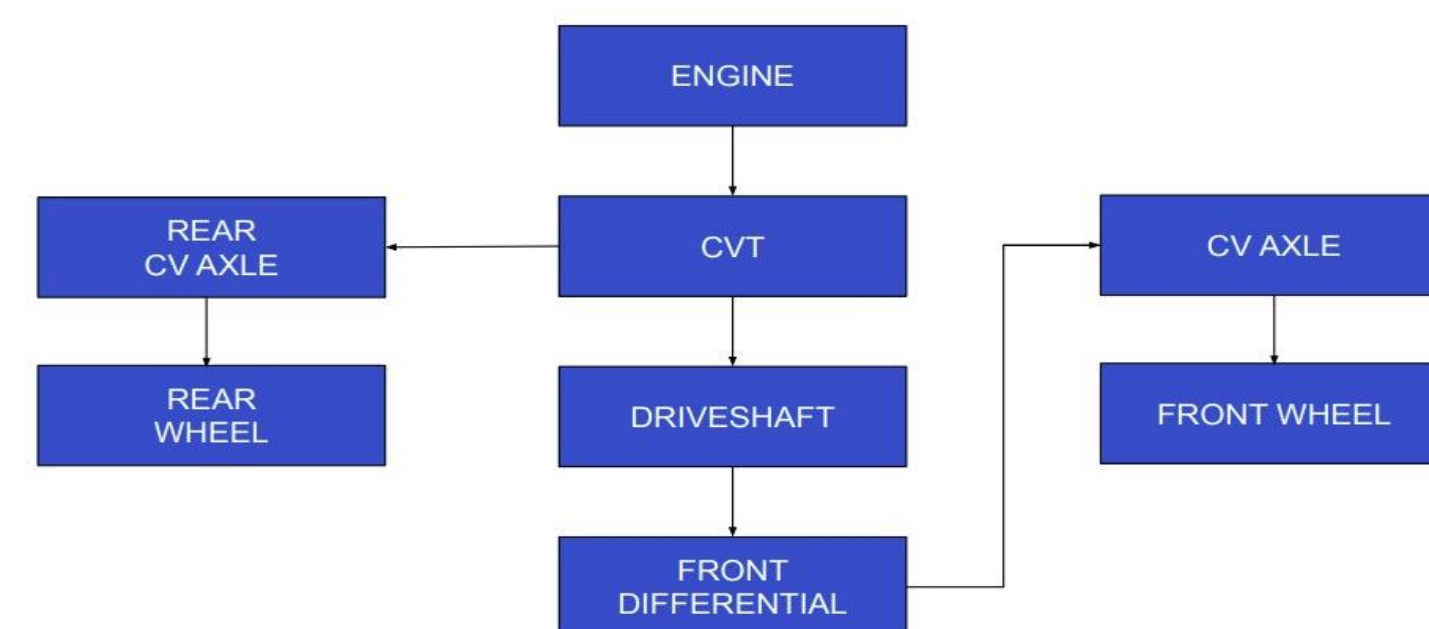
Front Double Wishbone A-Arm Design (CAD model and manufactured)



Rear Trailing Arm Design (CAD model and manufactured)

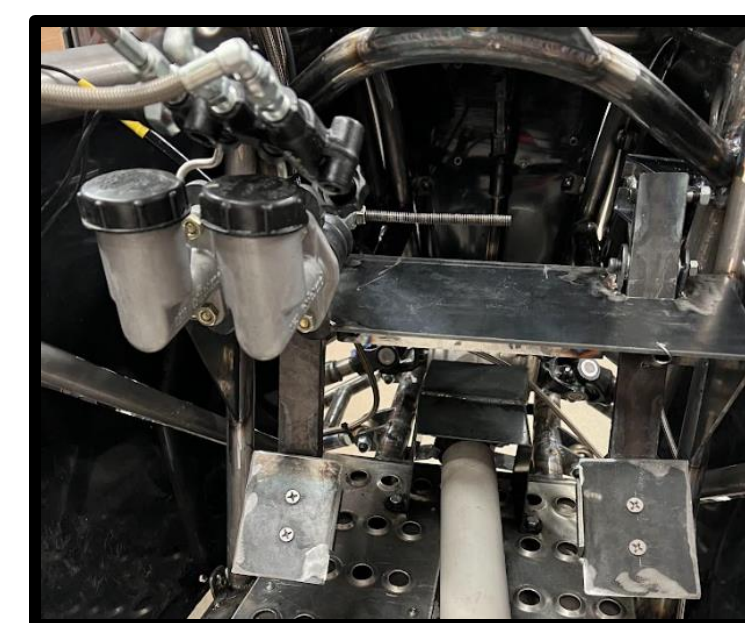
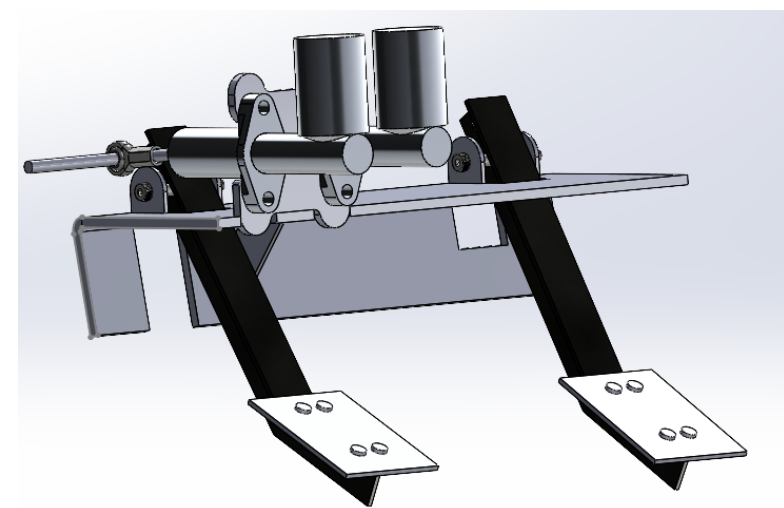
Drivetrain and Engine:

- CVT from a 2004 Polaris Sportsman 500 HO
- Low and high range gearing
- Forward and reverse
- Kohler CH440-3302 14 HP Command Pro Engine



Pedals (Throttle and Brake):

- 8:1 pedal ratio
- Wilwood master cylinders and proportioning valve



Constraints/Requirements

- Design and manufacture Western's first SAE Baja vehicle
- Finish manufacturing in time to attend the California competition in April 2024
- Abide by all the SAE rules allowing us to pass technical inspections
- Supply the upcoming senior class with a comprehensive vehicle and documentation to use as a platform for next year

Background

Baja SAE® consists of competitions that simulate real-world engineering design projects and their related challenges. Engineering students are tasked to design and build an off-road vehicle that will survive the severe punishment of rough terrain. Each team's goal is to design and build a single-seat, all-terrain, sporting vehicle whose structure contains the driver. The vehicle is to be a prototype for a reliable, maintainable, ergonomic, and economic production vehicle which serves a recreational user market. - SAE International



SAE Baja Scored Events

Static Events

- Business Event RPF
- Roll cage Documentation Packet
- Virtual Business Presentation
- Competition Design Report (DRB)
- Online Cost Prototype
- In Person Design Presentation

Dynamic Events

- Maneuverability Course
- Rock Crawl Course
- Acceleration
- Hill Climb
- 4hr Endurance Race

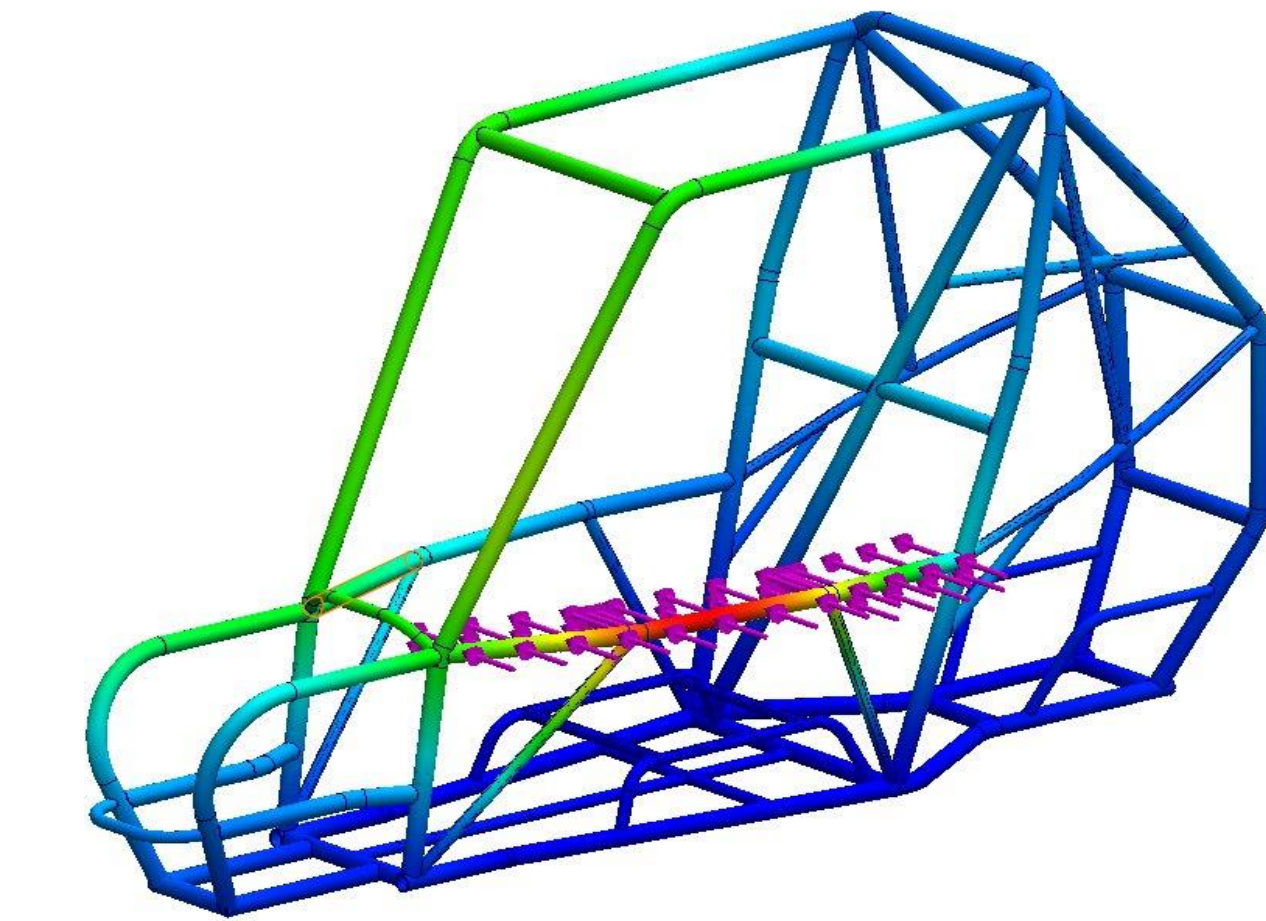
Technical Inspections

- Engine Check
- Frame Check
- Dynamic Brake Test
- 4WD Check
- General Tech Inspection

Future Work

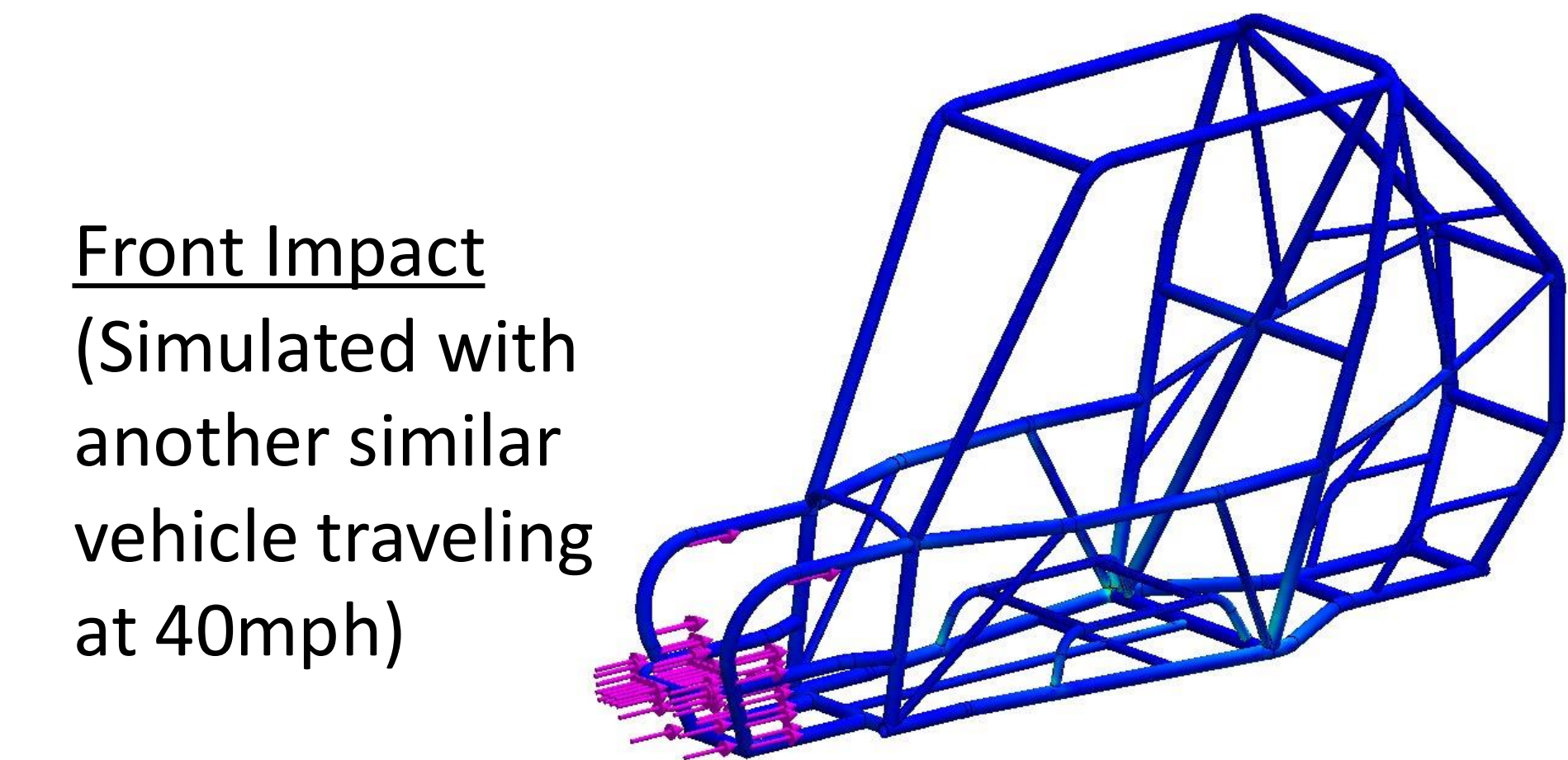
- Complete redesign of the chassis and roll cage around a tested and optimized suspension platform
- Reduce the weight and complexity of the car as a whole to increase speed and performance
- Future teams will design the car in one year and then go to competition the next

Testing



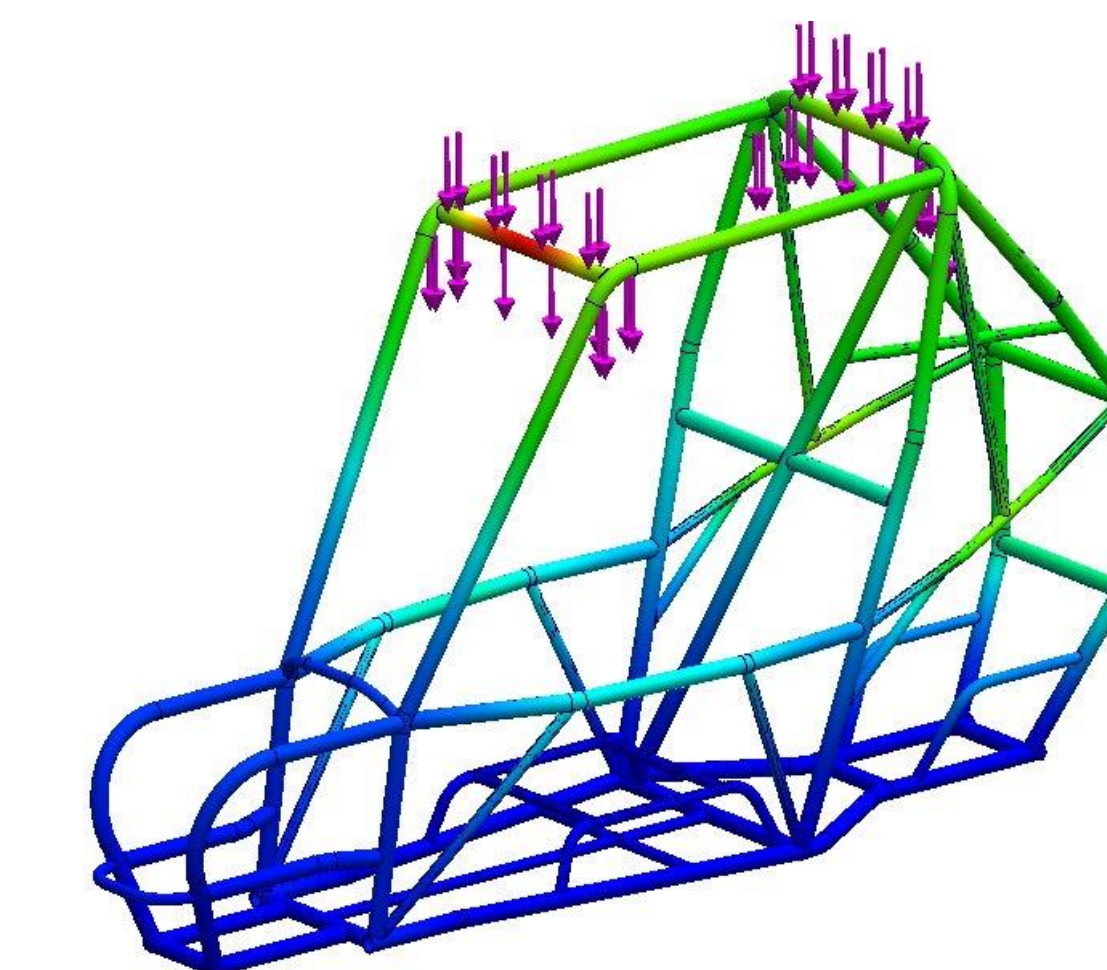
URES (mm)
2.752e+01
2.477e+01
2.202e+01
1.927e+01
1.651e+01
1.376e+01
1.101e+01
8.257e+00
5.505e+00
2.752e+00
1.000e+00

Side Impact
(Simulated with another similar vehicle traveling at 20mph)



Front Impact
(Simulated with another similar vehicle traveling at 40mph)

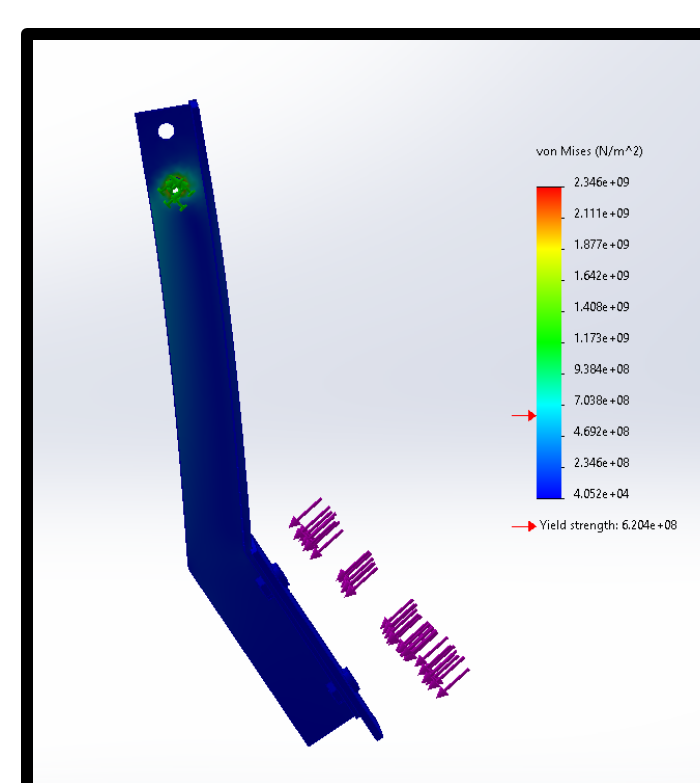
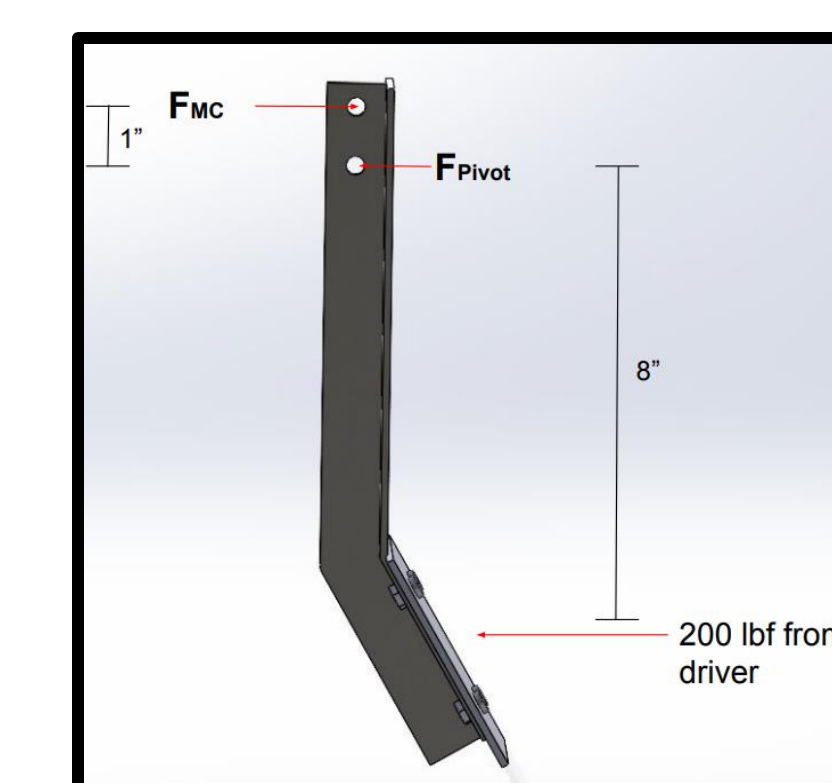
von Mises (N/mm²)
2.734e+10
2.461e+10
2.187e+10
1.914e+10
1.641e+10
1.367e+10
1.094e+10
8.203e+09
5.468e+09
2.734e+09
0.000e+00



URES (mm)
7.012e+01
6.311e+01
5.610e+01
4.909e+01
4.207e+01
3.506e+01
2.805e+01
2.104e+01
1.402e+01
7.012e+00
1.000e+00

Rollover
(Simulated with a gross vehicle weight of 800lbs dropped from 2 ft)

FEA simulation with a 400lb input force from the driver per SAE rules



- Mock technical inspections were performed before going to the competition
- The powertrain (2WD, 4WD, High and Low range) and suspension were tested and tuned by driving up and down steep inclines, boulders, and railroad ties

Results

- The vehicle will be raced at the SAE Baja competition in California April 2024